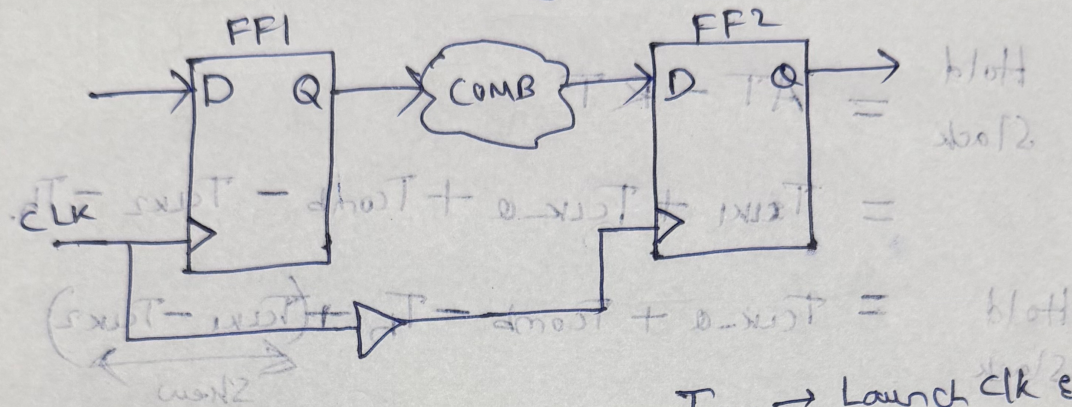


Practice Questions 01-STA

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1. STA - Static Timing Analysis



$$T_{clk} = 10ns$$

$$T_{clk1} - T_{clk2} = 0.2ns$$

$$T_{su} = 0.8ns$$

$$T_h = 0.4ns$$

$$T_{comb} = 7.2ns$$

$$T_{clk-q} = 0.5ns$$

$$a) AT = T_{clk-q} + T_{comb} + T_{clk1}$$

$$RT = T_{clk2} + T_{clk} - T_{su}$$

$$Setup Slack = RT - AT$$

$$= T_{clk} - T_{su} + T_{clk2} - (T_{clk1} + T_{comb} + T_{clk-q})$$

$$= 10 - 0.8 + 0.2 - 7.2 - 0.5$$

$$Setup Slack = 1.7ns$$

$$F_{max} = \frac{1}{T_{clk}} = \frac{1}{10ns} = 100MHz$$

b) Hold slack

$$AT = T_{clk1} + T_{clk0} + T_{comb}$$

$$RT = T_{clk2} + (Th)_{FP2} - AT$$

$$\begin{aligned} \text{Hold Slack} &= AT - RT \\ &= T_{clk1} + T_{clk0} + T_{comb} - T_{clk2} - Th \end{aligned}$$

$$\text{Hold Slack} = T_{clk0} + T_{comb} - Th + (T_{clk1} - T_{clk2})$$

← Skew →

$$= 0.5 + 7.2 - 0.4 - 0.2$$

$$\text{Hold Slack} = 7.1ns$$

c) The Setup & Hold Slack are positive only
Yes it meets the Timing Requirements.

d) Setup slack = 0

$$RT = AT$$

$$T_{clk2} + T_{clk} - T_{su} = T_{clk0} + T_{comb} + T_{clk1}$$

$$T_{clk} = T_{clk0} + T_{comb} + (T_{clk1} - T_{clk2}) + T_{su}$$

$$= 0.5 + 7.2 + (0.2) + 0.581 = 8.481ns$$

$$T_{clk} = 8.3ns$$

$$Freq = \frac{1}{T_{clk}} = 0.1204GHz \approx 120.5MHz$$

2. Digital Circuit contains 4 R-2-R paths with
Combinational delays.

Path Delay

P1 3.8ns

P2 5.6ns

P3 7.9ns

P4 6.4ns

$T_{clk} = 10ns$ $T_{su} = 0.8ns$

$T_{clk-a} = 0.5ns$

$$a) T + \text{Setup Slack} \leq RT - AT$$

$$AT = T_{clk-a} + T_{comb} + T_{clk}$$

$$RT = T_{clk2} + T_{clk} - T_{su}$$

$$\text{Setup Slack} = RT - AT$$

$$= T_{clk} - T_{su} + T_{skew} - T_{comb} - T_{clk-a}$$

Path	Setup Slack	Slack Values
P1	$10 - 0.8 + 0 - 3.8 - 0.5 = 4.9ns$	
P2	$10 - 0.8 + 0 - 5.6 - 0.5 = 3.1ns$	
P3	$10 - 0.8 + 0 - 7.9 - 0.5 = 0.8ns$	
P4	$10 - 0.8 + 0 - 6.4 - 0.5 = 2.3ns$	

b) Identify critical path.



Maximum Propagation delay path
How FF1 & FF2

Path 3 with $T_{comb} = 7.9 ns$

c) There exists no Timing Violations w.r.t to
based on Setup Slack values.

d) Maximum Operating frequency of design.

$$F_{max} = \frac{1}{T_{min}}$$

Path Timing Value $T \geq T_{clkq} + T_{comb} + T_{su}$

P1 $T \geq 0.5 + 3.8 + 0.8 = 5.1 ns$

P2 $T \geq 0.5 + 5.6 + 0.8 = 6.9 ns$

P3 $T \geq 0.5 + 7.9 + 0.8 = 9.2 ns$

P4 $T \geq 0.5 + 6.4 + 0.8 = 7.7 ns$

$$F_{max} = \frac{1}{5.1 ns} = 0.19607 GHz$$

$$F_{max} = 196 MHz$$

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3. Synchronous digital block =

$$F = 250 \text{ MHz}$$

$$\text{input delay} = 1.8 \text{ ns}$$

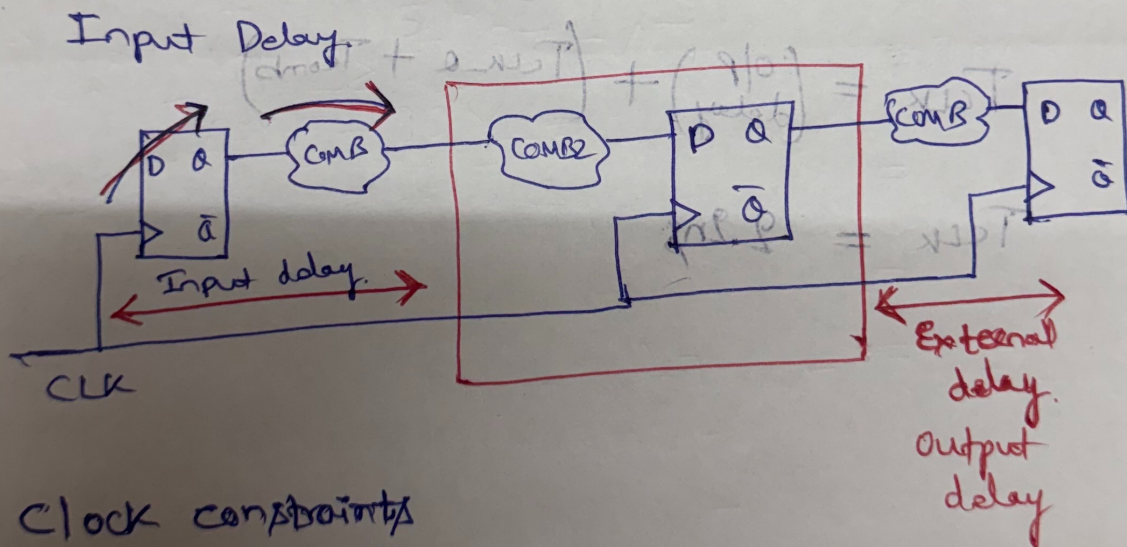
$$\text{output delay} = 2.2 \text{ ns}$$

a) Clock period for design.

Setup.

$$AT = (\text{input delay}) + t_{\text{comb2}} \Rightarrow 1.8 \text{ ns}$$

$$RT = T_{\text{clk}} - (t_{\text{out}})_{\text{FF2}} =$$



Clock constraints

1. Clock period.
2. Clock Uncertainty.

Setup time

Hold time

Not provided at all.

$$\text{Setup Slack} = RT - AT$$

$$= T_{CLK} - (T_{su})_{FF2} \rightarrow (i/p \text{ delay}) = 4ns$$

250MHz
↓
1/4 = 4ns

$$= 0$$

$$T_{CLK} = (T_{su})_{FF2} + (i/p \text{ delay}) = 1.8ns$$

808.1 = setup time

~~Hold Slack = AT - RT~~

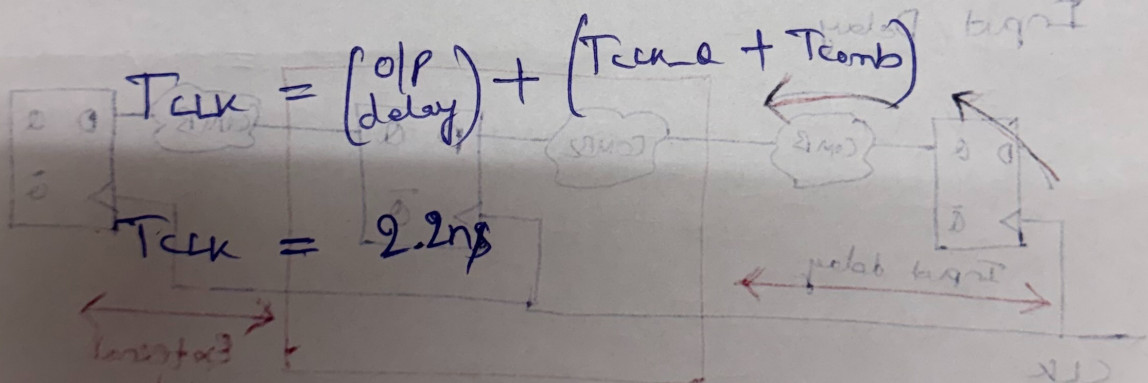
o/p delay

~~Requires~~

$$808.1 \leftarrow \text{setup} + (i/p \text{ delay}) = TA$$

$$\text{Setup Slack} = RT - AT$$

$$= T_{CLK} - (o/p \text{ delay}) - (T_{comb} + T_{comb})$$



Clock delay

1. Clock delay

2. Clock uncertainty

to be used for